Supplementary Table S1

Extended comparison of structural utilisation and related design evidence

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This curated comparison accompanies Section 2.2 and Table 1 in the paper “Strength to Spare, Deflection at the Limit: Governing Utilisation in 1.4 Million Structural Member Calculations”. The 22 entries report different measures and may include related datasets; their samples and outcomes are not pooled. Appraisal or redesign results do not establish directly removable material. Study IDs link to the sources and provenance notes that follow.

GWP = global warming potential; LCA = life-cycle assessment; RC = reinforced concrete; PT = post-tensioned. Carbon scopes and comparison limits remain source-specific.

Panel A — Utilisation, reserve and reinforcement appraisals

ID and study	Country and structure	Sample and measure	Relevant finding	Comparison limit
S01 — Moynihan and Allwood (2014)	UK; steel-framed buildings	23 designs; 10,130 valid beams and 1,837 columns	Reported mean beam utilisation 0.40 (unweighted) and 0.54 (mass-weighted); mean column utilisation 0.49	Selected projects; one material; appraisal and weighting affect comparison
S02 — Dunant et al. (2018)	UK; steel-frame floor plates	30 cases: 27 real and 3 models; governing utilisation	36% estimated unused steel mass under appraisal assumptions	Floor plates only; mass estimate is not measured removable material
S03 — Sadek et al. (2006)	Kuwait; reinforced-concrete villas	Six villas; existing reinforcement compared with ACI redesign at unchanged concrete dimensions	Redesign identified excess reinforcement in the existing designs	Small selected sample; quantity comparison rather than recorded governing utilisation; primary abstract does not quantify the excess
S04 — Alnuaimi et al. (2015)	Oman; reinforced-concrete villas	31 villas; provided versus redesigned reinforcement	48.5% average excess relative to redesign requirement	Quantity comparison rather than governing utilisation; minor overall cost effect
S05 — Forster (2016), MSt dissertation abstract	Cambridge study; in-situ concrete frames; project locations not stated	Sample RC projects; beam utilisation and slab-thickness appraisal; counts not reported in abstract	Mean beam utilisation about 0.70; slabs about 0.85 utilised by thickness, with lower reinforcement utilisation	Abstract-level evidence; sample size and detailed checks unavailable; slab thickness is a different utilisation measure
S06 — Mei et al. (2024)	UK non-seismic context; existing concrete columns	Appraisal of overdesign factors; typical 1960s flats	32% minimum and 85% likely reserve under stated assumptions	Conditional reserve estimates; not a routine-calculation distribution

Panel B — Redesign, structural-system and optimisation comparisons

ID and study	Country and structure	Sample and measure	Relevant finding	Comparison limit
S07 — Drewniok et al. (2020)	UK Eurocode context; non-composite steel beams	Catalogue-based redesign with fixed layout	26.5% mass savings in original-constraint scenario	Conditional optimisation outcome, rather than routine-practice prevalence
S08 — Gillott et al. (2025)	UK code context; steel office frames	11,252 hypothetical configurations; 11 scenarios	Reserve and extension potential vary with design and appraisal	Generated cases do not establish observed utilisation frequencies
S09 — Miller et al. (2013)	Australia, South East Queensland; concrete office slabs	One 10-storey office; reinforced and post-tensioned slab alternatives; material quantities	Post-tensioned alternative used 36.9% less concrete volume and 43.4% less steel mass	Single design comparison of alternative slab systems; savings are conditional on the study design, rather than observed reserve
S10 — Dunn (2015), FWPA project PNA308-1213	Australia, Sydney reference location; timber versus concrete or steel	Four building types: office, apartment, aged care and industrial shed; structural cost plans	Timber structural costs lower by 12.4%, 2.2%, 13.9% and 9.4%, respectively	Industry costing of alternative systems at a specified location and date; measures cost, rather than utilisation or removable material
S11 — Helal et al. (2020)	Australia-based study; parametric tall-building systems	80 structural systems; finite-element design and hybrid life-cycle inventory analysis	Changing structural design methods changed embodied greenhouse-gas emissions by up to 22%	Modelled systems; sensitivity to design and appraisal assumptions, rather than observed member-utilisation frequencies

Panel B — Redesign, structural-system and optimisation comparisons (continued)

ID and study	Country and structure	Sample and measure	Relevant finding	Comparison limit
S12 — Hawkins et al. (2020)	UK research; thin-shell concrete floors versus flat slabs	Design comparisons across 6–18 m spans; equivalent structural performance	Thin-shell floors reduced embodied carbon by 53–58% and self-weight by 43–53%	Conditional floor-system redesign; requires a different structural form and construction method; no prevalence estimate
S13 — Jayasinghe et al. (2022)	UK office-floor scenarios; concrete floor systems	Parametric spans and floor types; cradle-to-gate embodied carbon versus conventional flat slabs	Savings up to 12% from depth/grade, 23% from post-tensioning, 36% from alternative conventional slabs and 65% from thin shells	Strategy-specific scenario maxima; percentages cannot be added or read as observed unused capacity
S14 — Robati and Oldfield (2022)	Australia; mid-rise mass-timber and post-tensioned concrete buildings	Two reference buildings; Monte Carlo uncertainty in life-cycle embodied carbon	Timber 196–590 kgCO2e/m² (mean 417); concrete 307–618 (mean 465); mean difference 48 kgCO2e/m²	Whole-building carbon under inventory and life-cycle assumptions; overlapping ranges; no member reserve or routine-practice distribution
S15 — Huang et al. (2023)	Australia-based study; lightweight residential frame	One archetype; timber, steel and timber–steel composite frame options assessed using finite-element models	Embodied carbon 35.56% lower for timber and 8.12% lower for timber–steel composite than for steel	Single-archetype material and system comparison; embodied-carbon differences are not governing utilisation ratios
S16 — Feickert and Mueller (2025)	US design and policy context; steel and concrete buildings	Parametric building configurations; modelled embodied-carbon savings from policy and design strategies	Buy Clean savings about 9% (steel) and 16% (concrete); span reduction or shape-optimised floors yielded up to about 70% and 79%	Modelled, strategy-dependent savings against specified baselines; no observed utilisation sample or directly removable-material estimate

Panel C — Multi-project material-quantity and embodied-carbon datasets

ID and study	Country and structure	Sample and measure	Relevant finding	Comparison limit
S17 — Fang et al. (2026)	US and Canada; steel and reinforced concrete	226 designed systems from one firm; quantities and carbon	Component-level evidence on material and emissions	Does not measure governing utilisation or variation across firms
S18 — De Wolf et al. (2016)	Industry building survey; country split not reported	200 recently completed buildings; structural material mass and cradle-to-gate embodied CO2 per floor area	Structural quantities 200–1,800 kg/m²; embodied CO2 150–600 kgCO2e/m²	Multi-firm quantity and carbon evidence; no member demand-to-capacity ratios; country coverage cannot be inferred from affiliations
S19 — Simonen et al. (2017), CLF benchmark study	Worldwide compilation; structure, foundation and enclosure	Over 1,000 buildings; compiled initial embodied-carbon intensities (LCA stage A)	Initial embodied carbon typically below 1,000 kgCO2e/m²; 200–500 kgCO2e/m² covered 50% of commercial offices	Non-representative compilation with unaligned LCA methods and possible overlap with contributing datasets; carbon, rather than utilisation
S20 — Fang et al. (2024), SE 2050 executive summary	Primarily US; mixed building structural systems	522 submissions; filtered upfront-carbon n = 241 and A-to-C n = 100; structural GWP intensity	350 kgCO2e/m² proposed as an 80th-percentile upper limit for upfront structural GWP intensity, excluding biogenic carbon	Submitted-project carbon benchmark; subsets must be distinguished; spread limits precise system-specific benchmarks; no governing utilisation
S21 — Benke et al. (2024), California Carbon Report	US, California; structure and enclosure	30 new buildings from CLF WBLCA Benchmark Study v2; 60-year carbon assessment	Median embodied-carbon intensity of structure and enclosure 390 kgCO2e/m²	Regional, scope-limited LCA sample from the same CLF v2 programme; do not sum with the 292-project dataset; no utilisation ratios
S22 — Benke et al. (2025), dataset and companion CLF benchmark report	US and Canada; real building projects	292 projects; crowd-sourced design-practice LCA models with harmonised inventories and impacts	Open, detailed building-LCA dataset supporting embodied-carbon comparisons and benchmarks	Two publications from one dataset, counted once; building-LCA scope and possible overlap with other datasets preclude pooling; no governing utilisation

Sources and provenance notes

Evidence entries were reviewed on 26 October 2026, with bibliographic consistency updates on 28 October 2026. Access and verification limits are stated individually. These notes and the selection section form part of Table S1. Original sources should be cited when using their findings.

S01 — Moynihan and Allwood (2014)

Moynihan, M. C., and Allwood, J. M. (2014). Utilization of structural steel in buildings. Proceedings of the Royal Society A, 470(2168), 20140170.

DOI: 10.1098/rspa.2014.0170

Evidence access and verification: Primary open text checked for retained counts and unweighted versus mass-weighted means.

Dataset relationship: Selected designs from three consultancies; no pooled comparison is made.

Main Table 1: yes. Main Section 2.2 prose: yes.

S02 — Dunant et al. (2018)

Dunant, C. F., Drewniok, M. P., Eleftheriadis, S., Cullen, J. M., and Allwood, J. M. (2018). Regularity and optimisation practice in steel structural frames in real design cases. Resources, Conservation and Recycling, 134, 294–302.

DOI: 10.1016/j.resconrec.2018.01.009

Evidence access and verification: Comparator retained in the paper's Table 1; governing checks and conditional unused-mass appraisal retained.

Dataset relationship: Dunant et al. (2021) reuses 19 designs from this floor-plate database; no separate entry is added.

Main Table 1: yes. Main Section 2.2 prose: yes.

S03 — Sadek et al. (2006)

Sadek, A. W., Al-Fadala, S., and Al-Mutairi, N. (2006). Improving the design of residential buildings in Kuwait. Emirates Journal for Engineering Research, 11(2), 59–65.

Source: primary paper copy in CiteSeerX (Emirates Journal for Engineering Research, 2006)

Evidence access and verification: Primary abstract confirms six villas and overdesign. The numerical excess figures quoted by Alnuaimi were not independently verified in the primary full text and are excluded.

Dataset relationship: Predecessor of the Oman redesign study; separate country/sample, not a replicated utilisation distribution.

Main Table 1: no. Main Section 2.2 prose: no.

S04 — Alnuaimi et al. (2015)

Alnuaimi, A. S., Al Mohsin, M., Hago, A., and El Gamal, S. (2015). Overdesign of villa structures in Oman. The Journal of Engineering Research, 12(2), 68–76.

DOI: 10.24200/tjer.vol12iss2pp68-76

Evidence access and verification: Publisher abstract checked for 31 villas, 48.5% reinforcement excess, and minor overall cost effect. Publisher citation supplies pages 68–76.

Dataset relationship: Distinct from Sadek et al. (2006); reinforcement redesign outcome.

Main Table 1: yes. Main Section 2.2 prose: yes.

S05 — Forster (2016), MSt dissertation abstract

Forster, P. (2016, October). Quantifying the opportunity to reduce material use in insitu reinforced concrete frames. Construction Engineering Masters dissertation abstract, University of Cambridge.

Source: University of Cambridge dissertation abstract PDF

Evidence access and verification: One-page primary abstract only. Signature reads Peter Forster, October 2016. Sample/member counts and full appraisal methods are unavailable; none are inferred.

Dataset relationship: Beam, slab-thickness and reinforcement utilisation are distinct measures within the same dissertation.

Main Table 1: no. Main Section 2.2 prose: yes.

S06 — Mei et al. (2024)

Mei, X., Hawkins, W., Darby, A., and Ibell, T. (2024). Realising the reserve capacity in existing reinforced concrete columns. Structures, 63, 106360.

DOI: 10.1016/j.istruc.2024.106360

Evidence access and verification: Primary university publication record/abstract checked for conditional minimum 32% and likely 85% reserve.

Dataset relationship: Existing-building appraisal; no observed routine-design sample is inferred.

Main Table 1: yes. Main Section 2.2 prose: yes.

S07 — Drewniok et al. (2020)

Drewniok, M. P., Campbell, J., and Orr, J. (2020). The Lightest Beam Method – A methodology to find ultimate steel savings and reduce embodied carbon in steel framed buildings. Structures, 27, 687–701.

DOI: 10.1016/j.istruc.2020.06.015

Evidence access and verification: Comparator retained in the paper’s Table 1; retain the original-constraint scenario for the 26.5% mass saving.

Dataset relationship: Conditional catalogue redesign; no additional observed practice population.

Main Table 1: yes. Main Section 2.2 prose: yes.

S08 — Gillott et al. (2025)

Gillott, C., Davison, B., and Densley Tingley, D. (2025). Reserve capacity and vertical extension potential in steel-frame buildings. Proceedings of the Institution of Civil Engineers – Structures and Buildings, 178(3), 235–247.

DOI: 10.1680/jstbu.24.00010

Evidence access and verification: Comparator retained in the paper's Table 1; 11,252 generated configurations and 11 scenarios retained.

Dataset relationship: Hypothetical cases must not be combined with counts of observed projects or members.

Main Table 1: yes. Main Section 2.2 prose: yes.

S09 — Miller et al. (2013)

Miller, D., Doh, J.-H., Guan, H., Mulvey, M., Fragomeni, S., McCarthy, T., and Peters, T. (2013). Environmental impact assessment of post tensioned and reinforced concrete slab construction. In From Materials to Structures: Advancement through Innovation. CRC Press.

DOI: 10.1201/b15320-180

Evidence access and verification: Chapter text checked in the source review for the 10-storey reference office and 36.9% concrete/43.4% steel differences. Publication year and the seven-author list were reconciled against the Griffith University repository record on 28 October 2026; ACMSM 2012 is the conference name.

Dataset relationship: Two slab-system alternatives for one reference office, not two independent buildings.

Main Table 1: no. Main Section 2.2 prose: yes.

S10 — Dunn (2015), FWPA project PNA308-1213

Dunn, A. (2015, January). Final report for commercial building costing case studies – Traditional design versus timber project. Forest and Wood Products Australia, project PNA308-1213.

Source: FWPA report PDF, project PNA308-1213

Evidence access and verification: Primary report checked for Andrew Dunn, January 2015, project identifier and the four structural-cost comparisons.

Dataset relationship: Four costed archetypes at a reference location/date; alternative systems are paired comparisons.

Main Table 1: no. Main Section 2.2 prose: no.

S11 — Helal et al. (2020)

Helal, J., Stephan, A., and Crawford, R. H. (2020). The influence of structural design methods on the embodied greenhouse gas emissions of structural systems for tall buildings. Structures, 24, 650–665.

DOI: 10.1016/j.istruc.2020.01.026

Evidence access and verification: Primary abstract/paper text checked for 80 systems and up to 22% sensitivity to design method.

Dataset relationship: The separate IOP paper on life-cycle inventory methods is not another utilisation dataset.

Main Table 1: no. Main Section 2.2 prose: yes.

S12 — Hawkins et al. (2020)

Hawkins, W., Orr, J., Ibell, T., and Shepherd, P. (2020). A design methodology to reduce the embodied carbon of concrete buildings using thin-shell floors. Engineering Structures, 110195.

DOI: 10.1016/j.engstruct.2020.110195

Evidence access and verification: Primary Cambridge repository abstract checked for spans, carbon and self-weight comparisons.

Dataset relationship: Floor-form strategy evidence; related thin-shell scenarios also occur in Jayasinghe et al. (2022).

Main Table 1: no. Main Section 2.2 prose: no.

S13 — Jayasinghe et al. (2022)

Jayasinghe, A., Orr, J., Hawkins, W., Ibell, T., and Boshoff, W. P. (2022). Comparing different strategies of minimising embodied carbon in concrete floors. Journal of Cleaner Production, 131177.

DOI: 10.1016/j.jclepro.2022.131177

Evidence access and verification: Primary abstract/results text checked for strategy-specific maxima. The maxima are not additive.

Dataset relationship: Overlapping strategy family with thin-shell research; entries are publications, not independent pooled observations.

Main Table 1: no. Main Section 2.2 prose: no.

S14 — Robati and Oldfield (2022)

Robati, M., and Oldfield, P. (2022). The embodied carbon of mass timber and concrete buildings in Australia: An uncertainty analysis. Building and Environment, 214, 108944.

DOI: 10.1016/j.buildenv.2022.108944

Evidence access and verification: Primary repository abstract checked in the expanded-table review for two reference buildings, ranges and means; uncertainty scope retained.

Dataset relationship: Monte Carlo outcomes from two reference buildings; ranges are not a sample of observed member reserve.

Main Table 1: no. Main Section 2.2 prose: no.

S15 — Huang et al. (2023)

Huang, B., Xing, K., and Rameezdeen, R. (2023). Exploring embodied carbon comparison in lightweight building structure frames: A case study. Sustainability, 15(20), 15167.

DOI: 10.3390/su152015167

Evidence access and verification: Primary article text checked for 35.56% timber and 8.12% hybrid reductions relative to steel.

Dataset relationship: Three frame options for one archetype, not three independent practice samples.

Main Table 1: no. Main Section 2.2 prose: no.

S16 — Feickert and Mueller (2025)

Feickert, K., and Mueller, C. T. (2025). Policy and design levers for minimizing embodied carbon in United States buildings: A quantitative comparison of current and proposed strategies. Building and Environment, 270, 112485.

DOI: 10.1016/j.buildenv.2024.112485

Evidence access and verification: Primary publisher abstract checked for approximate policy and design savings; scenario/baseline dependence retained.

Dataset relationship: Parametric policy/design comparisons; generated alternatives are not observed utilisation frequencies.

Main Table 1: no. Main Section 2.2 prose: no.

S17 — Fang et al. (2026)

Fang, D., Kenny, P., and Mueller, C. (2026). Material efficiency as a key opportunity to reduce embodied carbon in structural systems: Data insights from 226 fully designed projects. Journal of Structural Engineering, 152(2), 04025275.

DOI: 10.1061/JSENDH.STENG-14451

Evidence access and verification: Comparator retained in the paper's Table 1; source metadata taken from the reference record; quantities/carbon measure distinguished from D/C.

Dataset relationship: One-firm programme overlaps the earlier Thornton Tomasetti tracking programme. Possible overlap with other project databases is not resolved; counts are not summed.

Main Table 1: yes. Main Section 2.2 prose: yes.

S18 — De Wolf et al. (2016)

De Wolf, C., Yang, F., Cox, D., Charlson, A., Hattan, A. S., and Ochsendorf, J. (2016). Material quantities and embodied carbon dioxide in structures. Proceedings of the Institution of Civil Engineers – Engineering Sustainability, 169(4), 150–161.

DOI: 10.1680/ensu.15.00033

Evidence access and verification: Primary Crossref abstract checked for 200 buildings and quantity/carbon ranges. Country distribution is not reported.

Dataset relationship: Contributing industry datasets may recur in later compilations, including CLF; no pooling is justified.

Main Table 1: no. Main Section 2.2 prose: no.

S19 — Simonen et al. (2017), CLF benchmark study

Simonen, K., Rodriguez, B., McDade, E., and Strain, L. (2017). Embodied carbon benchmark study: LCA for low carbon construction. Carbon Leadership Forum, University of Washington.

https://hdl.handle.net/1773/38017

Evidence access and verification: Primary report checked in the expanded-table review for broad initial-carbon ranges and non-representativeness. Report authors follow its suggested citation, distinguished from the associated database authors.

Dataset relationship: Includes contributed datasets, including MIT deQo/Thornton Tomasetti. Methods and scopes are unaligned and overlap is possible.

Main Table 1: no. Main Section 2.2 prose: no.

S20 — Fang et al. (2024), SE 2050 executive summary

Fang, D., Chafart, M., Torres, M., and Broyles, J. (2024). Executive summary of the SE 2050 Commitment Program 2023 Data Analysis and Findings Report. Edited by L. Wingo, Z. Chabot, and E. Borchers. SE 2050.

https://se2050.org/wp-content/uploads/2024/10/2023-exec-summary-expanded-to-share.pdf

Evidence access and verification: Primary executive summary read for 522 submissions, upfront subset 241, A-to-C subset 100 and proposed 350 kgCO2e/m² upper limit; biogenic carbon excluded.

Dataset relationship: Subsets are not independent additional projects. Firm contributions may overlap other structural-carbon programmes; no combined total is reported.

Main Table 1: yes. Main Section 2.2 prose: yes.

S21 — Benke et al. (2024), California Carbon Report

Benke, B., Roberts, M., Shen, Y., Carlisle, S., Chafart, M., and Simonen, K. (2024). The California carbon report: An analysis of the embodied and operational carbon impacts of 30 buildings. Carbon Leadership Forum, University of Washington.

https://hdl.handle.net/1773/51287

Evidence access and verification: Primary CLF report/summary checked for 30 new buildings, 60-year scope and 390 kgCO2e/m² median for structure and enclosure.

Dataset relationship: Part of CLF WBLCA Benchmark Study v2. Do not add these 30 to the broader 292-project dataset.

Main Table 1: no. Main Section 2.2 prose: no.

S22 — Benke et al. (2025), dataset and companion CLF benchmark report

Benke, B., Chafart, M., Shen, Y., Ashtiani, M. Z., Carlisle, S., and Simonen, K. (2025). A harmonized dataset of high-resolution embodied life cycle assessment results for buildings in North America. Scientific Data. Companion: Benke, B., Jensen, A., Chafart, M., Simonen, K., and Lewis, M. (2025). The embodied carbon benchmark report: Embodied carbon budgets and analysis of 292 buildings in the US and Canada. Carbon Leadership Forum.

DOI: 10.1038/s41597-025-05216-0

DOI: 10.6084/m9.figshare.32827298

Evidence access and verification: Primary Scientific Data abstract checked for 292 projects and harmonisation. Companion report bibliographic identity checked on the CLF page. Unverified office percentiles are excluded.

Dataset relationship: Dataset paper and report describe the same 292-project programme and occupy one entry. Related California/industry inputs must not be added to it.

Main Table 1: no. Main Section 2.2 prose: no.

Search and selection note

Purpose and routes

This is a curated comparison used to position an empirical study of recorded governing utilisation. It is not a systematic review, a prevalence meta-analysis or a census of global practice. The seven comparators from the paper's initial literature review were supplemented through four Exa search passes on 26 October 2026: Australia/New Zealand; United States academic studies; United States industry and professional bodies; and global utilisation audits. Citation tracing and direct publisher, university-repository and report checks followed. Bibliographic details and source limits were reconciled when building this package.

The working search note records sources_reviewed: 378. That is a search-process count, not a reproducible count of unique deduplicated records or full texts screened. No PRISMA screening count, complete search-string archive or exhaustive coverage is claimed. The structured table states the access/verification boundary for each included entry. The original working note remains an internal record; it is superseded by S1 for the selected table content.

Inclusion and organisation

Retain studies that quantify member utilisation or conditional reserve; compare provided and redesigned reinforcement; or explain how redesign, structural-system choice, material quantity, cost and carbon relate to interpretation of a utilisation benchmark. Direct governing-utilisation evidence is distinguished from related outcomes. The 22 entries form three panels: utilisation/reserve/reinforcement (6); redesign/system/optimisation comparisons (10); and multi-project quantity/carbon datasets (6). Panel membership describes the principal role of an entry and is not a quality ranking.

Main-table selection

The eight main entries are S01, S02, S04, S06, S07, S08, S17 and S20. The selection retains the closest observed steel utilisation comparators; concrete reinforcement and conditional reserve; fixed-constraint redesign and generated-frame appraisal; and single-firm versus multi-firm project-level quantity/carbon evidence. This locates the remaining gap in measure and population coverage, rather than project count alone. Geography, sample size or a large saving percentage alone is insufficient for selection.

Forster (S05) remains explicitly acknowledged in the main prose because its concrete-utilisation evidence is closely related. It is outside the compact numerical table because only a one-page abstract is available and the sample counts and appraisal details are missing. Miller (S09) and Helal (S11) retain Australian context in a short synthesis sentence. All other entries remain accessible in S1; their placement does not imply that they are invalid.

Exclusions and related sources

• Dunant et al. (2021), DOI 10.1016/j.istruc.2021.04.033, reuses 19 designs from Dunant et al. (2018); it is not another independent empirical row.

• Webster/SEI (2020) and later professional summaries repeat Moynihan and Allwood (2014). Gholam's discussion repeats the Dunant sample. They are not separate US or other utilisation audits.

• Earlier Thornton Tomasetti reports on more than 600 structures belong to the tracking programme later analysed by Fang et al. (2026). They are not added as another independent population.

• Helal et al.'s IOP inventory-method comparison, DOI 10.1088/1755-1315/588/3/032028, measures inventory-method sensitivity rather than member reserve.

• Concrete-mixture strength overdesign, seismic overstrength factors, reliability calibration and comparisons between design-code sizing rules measure different constructs and are outside this comparison's utilisation scope.

- Practitioner surveys about target utilisation and reasons for reserve are contextual evidence, not observed member-utilisation distributions.
- Policy pages, EPDs, design guidance, national stock estimates and sources without a retrievable relevant result were not added as empirical study rows.

Verification and overlap limits

The qualitative Sadek entry excludes the secondary 30%, 60% and 240% figures because those figures were not verified in the primary full text. Forster has no inferred sample count. The unverified office percentiles from the CLF companion report are excluded. The CLF 292-project dataset paper and companion report are one entry; the California sample is part of the same programme. Other industry contributions may overlap CLF and SE 2050. Project, model, member and submission counts are kept distinct and are never summed.

No Australian or United States multi-project audit of routine governing member demand-to-capacity ratios was identified in this search. This describes the search outcome; it does not prove that none exists. The present manuscript's gap statement remains bounded to the reviewed evidence and its sampled platform population.